Github Link of the project: https://github.com/sunnysavita10/Finetuning-on-aws

Step 1: Create IAM Roles

We'll create two roles:

- SageMakerLLMRole
- LambdaInvokeLLMRole

You can write custom name

A. Create SageMaker role

a. Go to AWS Console → IAM → Roles → Create Role
b. Choose AWS Service → SageMaker
c. Attach policies:
AmazonS3FullAccess
AmazonSageMakerFullAccess
CloudWatchFullAccess

B. Create Lambda role
a. Go to AWS Console → IAM → Roles → Create Role
b. Choose AWS Service → Lambda
c. Attach policies:
AmazonSageMakerFullAccess
AmazonDynamoDBFullAccess
AmazonS3FullAccess
CloudWatchLogsFullAccess

STEP 2 — Create S3 Buckets for Dataset & Model Artifacts

1. First bucket for dataset storage
2. Second bucket for model artifacts (after training, SageMaker will save the model)

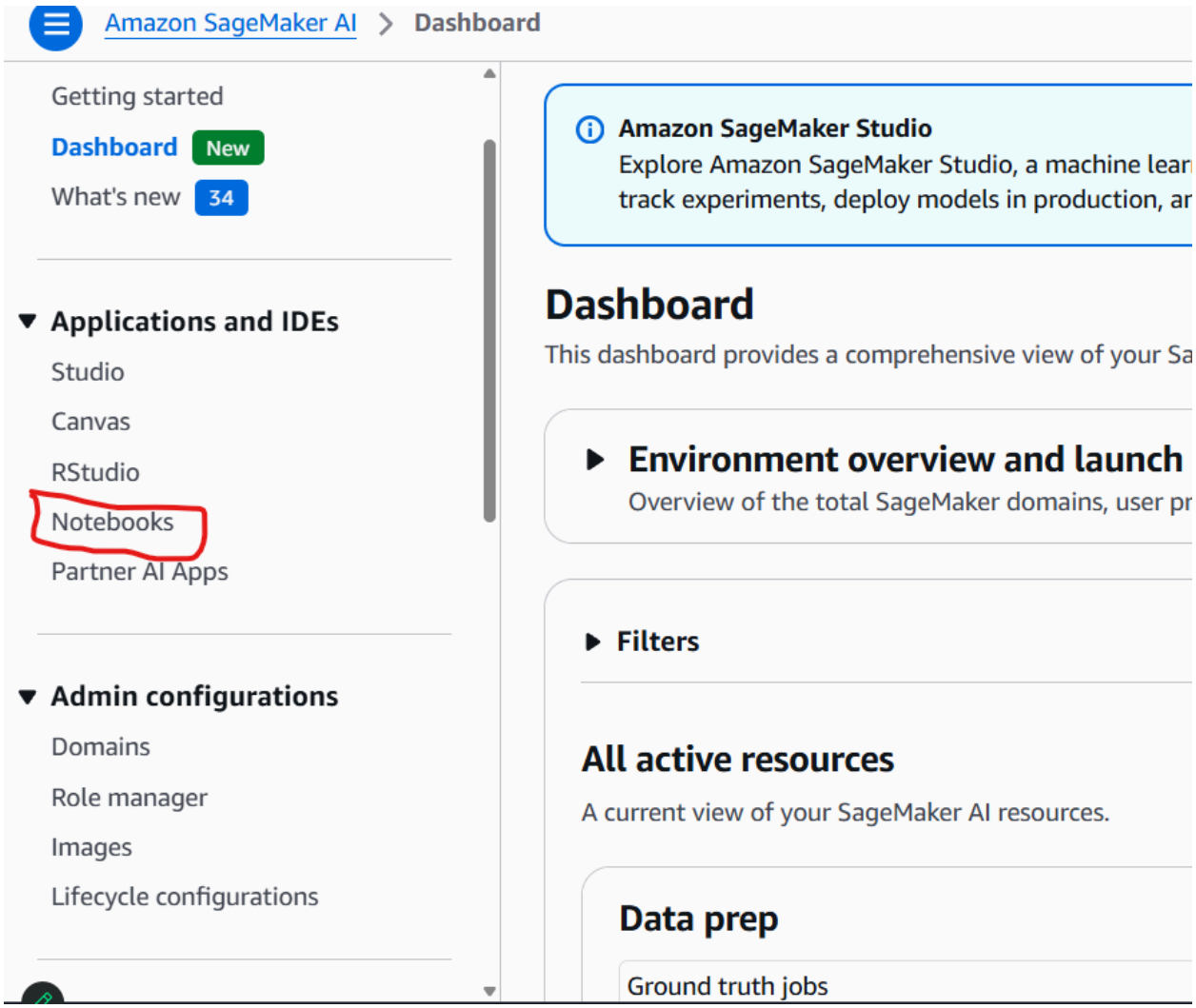
1. AWS Console → S3 → “Create bucket”
2. Bucket name: llm-finetune-dataset-<yourname>
3. Region = same as your SageMaker region (e.g., ap-south-1)
4. Create a bucket

Why have we created this bucket for keeping the training files

B. Create Second Bucket (for model artifacts)
1. AWS Console → S3 → “Create bucket”
2. Bucket name: llm-model-artifacts-<yourname>
3. Region = same as above
4. Create a bucket

C. Verify and Note the paths
s3://llm-finetune-dataset-sunny/datasets/
s3://llm-model-artifacts-sunny/models/

STEP 3 — Create SageMaker Notebook Instance



Amazon SageMaker AI > Notebook instances > Create notebook instance

Create notebook instance

Amazon SageMaker provides pre-built fully managed notebook instances that run Jupyter notebooks. The notebook instances include example code for common model training and hosting exercises. Learn more

Notebook instance settings

Notebook instance name

llm-finetune-notebook

Maximum of 63 alphanumeric characters. Can include hyphens (-), but not spaces. Must be unique within your account in an AWS Region.

Notebook instance type

ml.t3.medium

Platform identifier Learn more

Amazon Linux 2, Jupyter Lab 4

▸ **Additional configuration**

Permissions and encryption

IAM role

Notebook instances require permissions to call other services including SageMaker and S3. Choose a role or let us create a role with the AmazonSageMakerFullAccess IAM policy attached.

SageMakerLLMRole

Create role using the role creation wizard

Root access - *optional*

- Enable - Give users root access to the notebook
- Disable - Don't give users root access to the notebook

Lifecycle configurations always have root access

Encryption key - *optional*

Encrypt your notebook data. Choose an existing KMS key or enter a key's ARN.

No Custom Encryption

▸ **Network** - *optional*

▸ **Git repositories** - *optional*

▸ **Tags** - *optional*

Cancel | Create notebook instance

S3

Amazon SageMaker AI > Notebook instances > llm-finetune-notebook

Dashboard New
What's new 34

Applications and IDEs
Studio
Canvas
RStudio
Notebooks
Partner AI Apps

Admin configurations
Domains
Role manager
Images
Lifecycle configurations

Search

JumpStart
Foundation models
Computer vision models
Natural language processing models

Governance

HyperPod Clusters

llm-finetune-notebook

Delete | Stop | Open Jupyter | Open JupyterLab

Notebook instance settings

Edit

Name	Status	Notebook instance type	Platform identifier
llm-finetune-notebook	InService	ml.t3.medium	Amazon Linux 2, Jupyter Lab 4 (notebook-al2-v3)

ARN
arn:aws:sagemaker:ap-south-1:730335253621:notebook-instance/llm-finetune-notebook

Creation time
Nov 09, 2025 09:58 UTC

Volume Size
5GB EBS

Minimum IMDS Version
2

Lifecycle configuration
-

Last updated
Nov 09, 2025 10:00 UTC

Git repositories

Name	Repository URL	Type
There are currently no resources.		

Permissions and encryption

IAM role ARN
arn:aws:iam::730335253621:role/SageMakerLLMRole

Root access
Enabled

Encryption key

Network

No custom VPC settings applied

CloudShell | Feedback | Console Mobile App | You are screen sharing | Stop share

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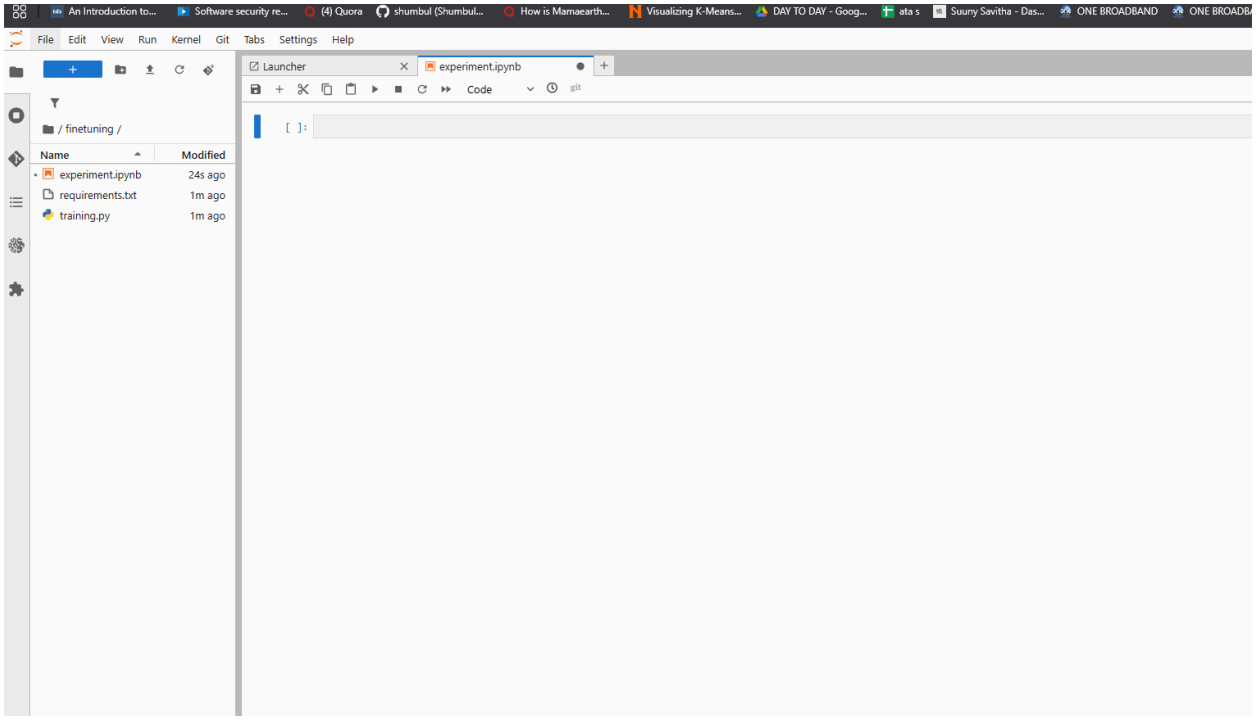
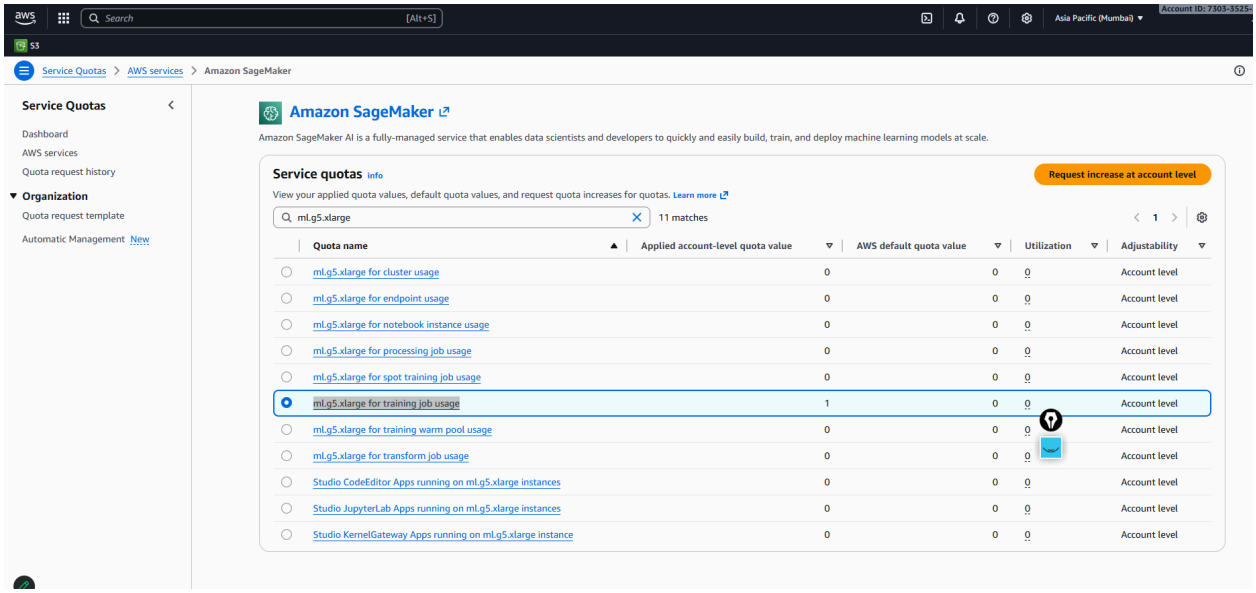


Keep the necessary requirements in the requirements.txt
Then install with the below command:
Pip install -r requirements.txt

Note: please check with your requirements.txt path

1. ls
2. pwd
3. cd /home
4. ls
5. cd ec2-user/
6. ls
7. cd SageMaker/
8. ls
9. cd Finetuning-on-aws/
10. ls
11. git status
12. Clear
13. Activate the env:
    source /home/ec2-user/anaconda3/bin/activate python3
14. pip install --upgrade pip setuptools wheel
15. pip install --upgrade cmake
16. sudo yum install -y gcc gcc-c++ make
17. pip install cmake==3.27.0 pyarrow==16.1.0 --no-cache-dir
18. pip install -r ./finetuning/requirements.txt
19. pip cache purge

Open the page below to increase the quota



Create AWS Lambda Function

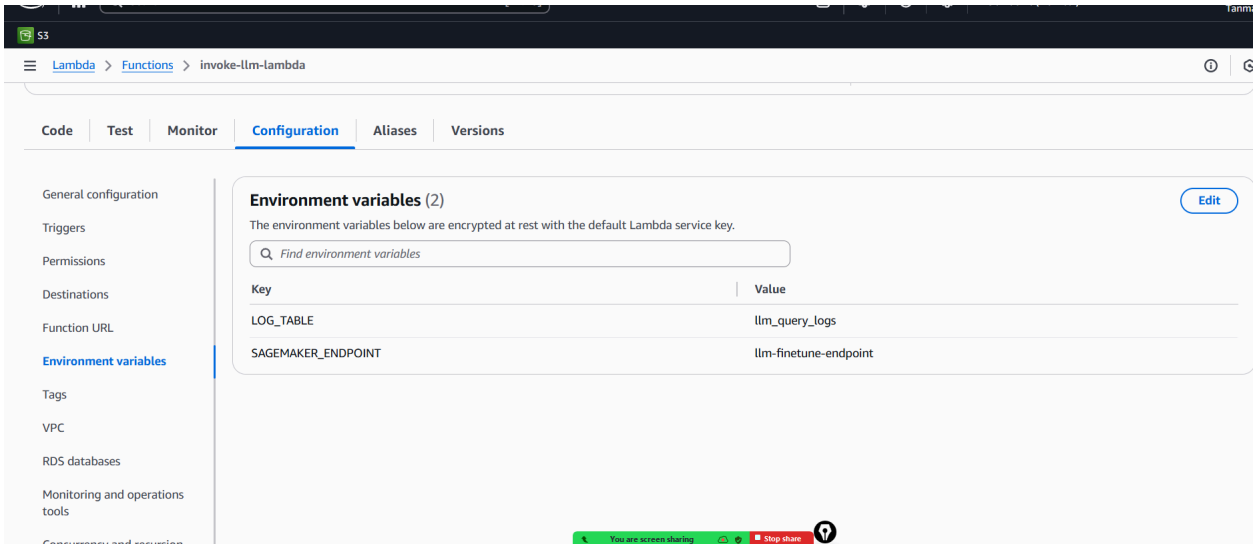
Go to **Lambda → Create function**

Name: `invoke-llm-lambda`

`Runtime: Python 3.10`

`Select Existing Role: LambdaInvokeLLMRole`

Add the env variable in the lambda



Paste handler:

```
import json, os, time, boto3


runtime = boto3.client("sagemaker-runtime")
dynamo = boto3.resource("dynamodb").Table(os.environ["LOG_TABLE"])
ENDPOINT = os.environ["SAGEMAKER_ENDPOINT"]

def lambda_handler(event, context):
    body = json.loads(event.get("body", "{}"))
    text = body.get("inputs", "")

    resp = runtime.invoke_endpoint(
        EndpointName=ENDPOINT,
```

```
        ContentType="application/json",
        Body=json.dumps({"inputs": text})
    )
    result = json.loads(resp["Body"].read().decode())

    log_item = {
        "id": f"{int(time.time()*1000)}#{context.aws_request_id}",
        "prompt": text,
        "response": result,
        "timestamp": int(time.time())
    }
    dynamo.put_item(Item=log_item)

    return {
        "statusCode": 200,
        "body": json.dumps({"result": result})
    }
```

Create DynamoDB Table (for logs)

Go to DynamoDB → Create table

- Table name: llm_queries
- Partition key: id (String)
- Sort key: (optional)

Deploy-test with the sample

```
{
  "body": "{\"inputs\": \"Summarize the text about heart disease\"}"
}
```

Create API Gateway (REST)

1. Go to API Gateway → Create API → REST API → Build
    - Name: LLMInferenceAPI
2. Create resource /predict
3. Create method POST
    - Integration type: Lambda Function
    - Choose invoke-llm-lambda
4. Deploy API → stage: prod

Now create a UI for the inferencing and RAG appp Create a virtual env and install the requirementS_inference.txt

1. uv python list
2. uv venv env --python cpython-3.11.13-windows-x86_64-none
3. env/Scripts/activate
4. source env/Scripts/activate→ for bash terminal
5. uv pip install -r requirements_inference.txt

Then after creating a app will run it via these below commands

6. streamlit run inference_app.py

Command for running Streamlit on a different port:
streamlit run rag_app_ui.py --server.port 8502

Env variables:

```
GROQ_API_KEY=""
GOOGLE_API_KEY=""
TAVILY_API_KEY=""
OPENAI_API_KEY=""
API_URL = ""  # Replace with your actual API Gateway endpoint
API_KEY= ""
```